

# AI Agent Trading Bot Platform

## Product Requirements Document (PRD)

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### 1. Document Information

| Field         | Value                                |
|---------------|--------------------------------------|
| Project Name  | AI Agent Trading Bot Platform        |
| Version       | 1.0                                  |
| Document Type | Product Requirements Document (PRD)  |
| Author        | Ayberk Yavuz                         |
| Date          | May 2026                             |
| Project Type  | AI Agent Based Crypto Trading System |
| Target Market | Binance Spot Market                  |

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### 2. Project Overview

The AI Agent Trading Bot Platform is an AI-powered automated crypto trading system designed to analyze market conditions, evaluate technical indicators and market sentiment, and perform short-term trading operations on Binance Spot Market.

The platform aims to:

- Provide practical AI agent engineering experience
- Learn AI agent orchestration and workflow design
- Learn cost-efficient LLM usage strategies
- Generate passive income through algorithmic trading
- Build a portfolio-quality AI engineering project
- Gain experience with modern AI infrastructure and production system design

The system will initially operate on a local Mac Mini environment and may later migrate to cloud infrastructure for 24/7 execution.

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## 3. Business Goals

### 3.1 Primary Goals

#### Goal 1 — AI Agent Engineering Experience

Gain hands-on experience with:

- AI agent workflows
- Tool usage
- Agent orchestration
- Model Context Protocol (future phase)
- LLM integrations
- AI observability
- Agent evaluation techniques
- Multi-agent system design
- Cost optimization techniques

#### Goal 2 — Cost-Efficient AI Architecture

Design a trading system where:

- Trading profits exceed LLM API costs
- LLM calls are minimized
- Deterministic systems handle repetitive tasks
- AI reasoning is selectively triggered

#### Goal 3 — Passive Income Generation

Build a short-term trading system capable of:

- Automated trading
- Real-time market analysis
- Buy/sell decision making
- Risk-controlled operation

#### Goal 4 — Portfolio Development

Create a production-quality project suitable for:

- LinkedIn portfolio
  - GitHub showcase
  - Technical interviews
  - AI engineering demonstrations
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## 4. Project Scope

### 4.1 Included Scope (Version 1)

The initial version of the system will support:

#### Trading Market

- Binance Spot Market only

#### Supported Assets

- BTC/USDT
- ETH/USDT
- Additional highly liquid crypto pairs may be added later

#### Core Features

- User authentication
- AI agent orchestration
- Market analysis
- Technical indicator analysis
- News and sentiment analysis
- Automated buy/sell execution
- Risk management
- Real-time transaction monitoring
- Live dashboard updates
- Trade history tracking
- Agent reasoning logs
- LLM cost monitoring

#### Deployment

- Local deployment on Mac Mini
  - Docker Compose based deployment
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### 4.2 Excluded Scope (Version 1)

The following features are intentionally excluded from the initial version:

#### Markets

- Turkish stock market
- US stock market
- Futures trading

- Options trading
- Forex trading

### **Advanced Trading Features**

- Leverage trading
- Margin trading
- High-frequency trading (HFT)
- Arbitrage trading
- Reinforcement learning
- Copy trading

### **Infrastructure**

- Kubernetes deployment
- Multi-region deployment
- Mobile application
- Multi-user SaaS support

### **AI Features**

- Autonomous recursive agents
  - Fully autonomous AGI trading
  - Long conversational memory
  - Self-modifying prompts
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## **5. Success Criteria**

### **5.1 Technical Success Criteria**

The system shall:

- Successfully connect to Binance APIs
- Execute automated buy/sell operations
- Display live transactions in the frontend
- Log all agent decisions into PostgreSQL
- Maintain stable operation on Mac Mini
- Provide observable agent workflows
- Maintain structured logs for debugging

### **5.2 AI Engineering Success Criteria**

The project shall demonstrate:

- LangGraph workflow orchestration

- AI tool calling
- Structured LLM outputs
- Agent workflow management
- Cost-efficient LLM usage
- Event-driven AI architecture

## 5.3 Financial Success Criteria

The system should aim to:

- Minimize unnecessary LLM API costs
  - Reduce bad trades using risk management
  - Operate with controlled capital exposure
  - Maintain profitable or near break-even operation during testing
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# 6. System Users

## 6.1 Primary User

The primary user is the system owner.

The user shall be able to:

- Login securely
  - Start trading agents
  - Stop trading agents
  - Monitor live trades
  - Monitor portfolio performance
  - Monitor AI decisions
  - Monitor LLM costs
  - View trade history
  - View agent reasoning logs
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# 7. Functional Requirements

## 7.1 Authentication Module

### FR-001 — User Login

The system shall allow users to login using email and password.

### **FR-002 — Session Management**

The system shall manage authenticated user sessions securely.

### **FR-003 — Logout**

The system shall allow users to logout securely.

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## **7.2 Trading Control Module**

### **FR-004 — Start Trading**

The user shall be able to start trading agents from the frontend dashboard.

### **FR-005 — Stop Trading**

The user shall be able to stop trading agents immediately.

### **FR-006 — Emergency Kill Switch**

The system shall support emergency shutdown of all active trading operations.

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## **7.3 Market Analysis Module**

### **FR-007 — Market Data Collection**

The system shall continuously collect market data from Binance APIs.

### **FR-008 — Technical Indicator Calculation**

The system shall calculate technical indicators including:

- RSI
- MACD
- Moving averages
- Volume spikes
- Volatility
- Momentum indicators

### **FR-009 — Signal Detection**

The system shall detect trading opportunities using deterministic rules.

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## 7.4 News & Sentiment Analysis Module

### FR-010 — News Retrieval

The system shall retrieve crypto-related market news.

### FR-011 — Sentiment Analysis

The system shall analyze market sentiment using LLM services.

### FR-012 — Structured AI Outputs

The system shall enforce JSON-based structured outputs from LLMs.

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## 7.5 Trade Decision Module

### FR-013 — Buy/Sell Decision Generation

The system shall generate BUY, SELL, or HOLD decisions.

### FR-014 — Confidence Score

The system shall generate confidence scores for each decision.

### FR-015 — Decision Logging

The system shall log all decisions and reasoning into PostgreSQL.

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## 7.6 Risk Management Module

### FR-016 — Position Size Validation

The system shall validate maximum position sizes before execution.

### FR-017 — Daily Loss Protection

The system shall stop trading after exceeding daily loss thresholds.

### FR-018 — Trading Frequency Control

The system shall limit excessive trading frequency.

### **FR-019 — Duplicate Trade Prevention**

The system shall prevent repetitive buy/sell loops.

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## **7.7 Trade Execution Module**

### **FR-020 — Binance Order Execution**

The system shall place market orders using Binance APIs.

### **FR-021 — Order Status Verification**

The system shall verify successful order execution.

### **FR-022 — Balance Verification**

The system shall validate available balances before trading.

### **FR-023 — Transaction Logging**

The system shall store all transactions in PostgreSQL.

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## **7.8 Dashboard Module**

### **FR-024 — Live Dashboard**

The system shall provide a live trading dashboard.

### **FR-025 — Real-Time Updates**

The system shall provide real-time frontend updates using WebSockets.

### **FR-026 — Portfolio Monitoring**

The system shall display:

- Current balance
- Open positions
- Daily profit/loss
- Historical performance



### **FR-027 — Agent Monitoring**

The dashboard shall display:

- Agent status
- Recent decisions
- Confidence scores
- Decision reasoning

### **FR-028 — LLM Cost Monitoring**

The system shall display:

- Token usage
  - API costs
  - Daily AI expenses
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## **8. Non-Functional Requirements**

### **8.1 Performance Requirements**

#### **NFR-001 — Low Latency**

The system should process trading decisions within acceptable short-term trading latency.

#### **NFR-002 — Real-Time Updates**

Frontend updates should appear with minimal delay.

#### **NFR-003 — Lightweight Architecture**

The system shall run efficiently on a Mac Mini environment.

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### **8.2 Reliability Requirements**

#### **NFR-004 — Failure Handling**

The system shall gracefully handle:

- Binance API failures
- LLM API failures
- Network interruptions
- Database connection issues

#### **NFR-005 — Safe Trading**

The system shall avoid uncontrolled trade execution.

#### **NFR-006 — Auditability**

All decisions and actions shall be traceable.

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### **8.3 Security Requirements**

#### **NFR-007 — API Key Security**

All API keys shall be securely stored.

#### **NFR-008 — Authentication Security**

Passwords shall be securely hashed.

#### **NFR-009 — Secure Communication**

Backend APIs shall use secure communication mechanisms.

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### **8.4 Maintainability Requirements**

#### **NFR-010 — Modular Architecture**

The system shall use modular architecture principles.

#### **NFR-011 — Structured Logging**

The system shall maintain structured JSON logs.

#### **NFR-012 — Containerized Deployment**

The system shall support Docker-based deployment.

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## 9. AI Agent Architecture Requirements

### 9.1 Market Scanner Agent

Responsibilities:

- Monitor Binance market data
- Detect technical signals
- Identify candidate trading opportunities

Characteristics:

- Primarily deterministic
  - Low-cost processing
  - Minimal LLM usage
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### 9.2 News & Sentiment Agent

Responsibilities:

- Analyze crypto-related news
- Evaluate market sentiment
- Interpret macro-level events

Characteristics:

- Uses LLM APIs
  - Structured JSON outputs
  - Event-driven execution
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### 9.3 Trade Decision Agent

Responsibilities:

- Generate BUY/SELL/HOLD decisions
- Combine technical and sentiment signals
- Produce confidence scores

Characteristics:

- LLM-assisted reasoning
- Structured outputs only
- Limited reasoning scope

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## 9.4 Risk Manager Agent

Responsibilities:

- Validate trading safety
- Enforce trading limits
- Prevent excessive losses

Characteristics:

- Deterministic rules
  - Minimal or no LLM usage
  - Highest system priority
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# 10. Cost Optimization Requirements

## 10.1 Event-Driven AI Usage

The system shall only trigger LLM reasoning during meaningful market events.

## 10.2 AI Cost Reduction

The system shall minimize unnecessary LLM calls.

## 10.3 Cached AI Responses

The system shall cache reusable AI outputs where applicable.

## 10.4 Structured Prompts

The system shall use compact structured prompts.

## 10.5 Lightweight Models

The system shall prioritize lightweight models for most operations.

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# 11. Technology Stack Requirements

## Frontend

- React
- Vite
- Tailwind CSS
- WebSocket support

## Backend

- FastAPI
- Python
- WebSockets
- SQLAlchemy
- Alembic

## Database

- PostgreSQL

## AI Agent Framework

- LangGraph
- Minimal LangChain usage

## Deployment

- Docker Compose

## Third-Party Services

- Binance API
- OpenAI API
- OpenRouter

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# 12. Initial Trading Constraints

## Initial Capital

- 200 USD

## Trading Type

- Short-term trading
- Low-frequency intelligent trades

## Risk Constraints

- No leverage
  - No margin trading
  - Limited simultaneous positions
  - Daily loss threshold
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# 13. Future Roadmap (Out of Scope for V1)

Potential future improvements may include:

- Multi-market trading
  - US stock market integration
  - Turkish stock market integration
  - Reinforcement learning
  - Cloud-native deployment
  - Kubernetes deployment
  - Mobile application
  - Multi-user SaaS architecture
  - MCP integrations
  - Advanced portfolio optimization
  - Backtesting engine
  - AI evaluation dashboards
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# 14. Assumptions

The project assumes:

- Binance APIs remain available
  - LLM APIs remain accessible
  - Initial trading volume remains low
  - System initially operates in a single-user environment
  - The project prioritizes learning and architecture development over immediate profitability
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# 15. Risks

## Technical Risks

- LLM hallucinations
- API downtime
- Incorrect market interpretation
- Unstable market conditions
- Token cost inflation

## Financial Risks

- Trading losses
- High LLM operational costs
- Unexpected market volatility

## Operational Risks

- Mac Mini downtime
  - Internet interruptions
  - Deployment failures
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# 16. Conclusion

The AI Agent Trading Bot Platform is designed as a modern AI engineering project focused on:

- AI agent orchestration
- Cost-efficient LLM usage
- Real-world trading automation
- Production-grade backend engineering
- Observability and explainability

The initial version intentionally limits scope to Binance Spot Market to prioritize:

- fast iteration
- lower operational complexity
- manageable AI costs
- improved system reliability
- practical AI engineering experience

The project architecture is designed for future scalability while maintaining a controlled and observable AI workflow-based system.